

SEIKH SOUVAGYA MUSTAKIM

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Education

Parala Maharaja Engineering College

Bachelor of Technology, Electronics and Telecommunications; CGPA: 8.39/10

2023 – 2027

Berhampur, Odisha

Nalanda Public School

Senior Secondary (Science); 91%

2020 – 2022

Cuttack, Odisha

St. Xavier's High School

Secondary; 93.8%

2010 – 2020

Athagarh, Odisha

Research & Internship Experience

IASc Summer Research Fellow

Dept. of Electronic Systems Engineering, IISc Bengaluru

- Selected for the highly competitive IASc-INSANA Summer Research Fellowship Programme (SRFP).
- Engineered a YOLO-based PCB reverse engineering pipeline (97%+ confidence) and custom OCR for BoM extraction; validated on FICS-PCB dataset on TrustHub.

Jun 2026 – Aug 2026

Bengaluru, India

Generative AI Internship

RMSoEE, IIT Kharagpur

- Completed a hands-on GenAI programme covering LLM fine-tuning, Retrieval-Augmented Generation (RAG), and Model Context Protocol (MCP); implemented end-to-end generative AI workflows.

Jun 2025 – Aug 2025

Technical Skills

- **Robotics & Autonomous Systems:** ROS2 Humble, Gazebo, RViz2, RPLidar, SLAM Toolbox, Nav2, AMCL, A* Path Planning, PID Control
- **Embedded Systems & Hardware:** Raspberry Pi, Arduino, ESP32, MPU6050, Jetson Nano, Tiva C (TM4C123).
- **Computer Vision & ML:** OpenCV, YOLOv8, MediaPipe, Lucas-Kanade Optical Flow, TensorRT, CNNs, OCR pipelines, Scikit-learn
- **Generative AI:** RAG pipelines, MCP, prompt engineering, vector databases
- **Programming & Tools:** Python, C, C++, Git, Linux, VS Code, Jupyter, CCS, PlatformIO

Projects

• Destination_bot — A* Path Planning on ROS2

[GitHub](#)

ROS2 · Gazebo · SLAM Toolbox · Nav2 · Python · RViz2

- Built a custom URDF differential-drive robot with a full SLAM pipeline and custom A* path planning node for autonomous collision-free navigation in Gazebo.

• RPLidar Visualisation & Gazebo Simulation

RPLidar A-Series · Pygame · Matplotlib · ROS2 · Gazebo

- Streamed real-time RPLidar scan data into 2D Pygame and 3D Matplotlib visualisations; integrated sensor into Gazebo + ROS2 as groundwork for SLAM.

• Face Recognition Attendance System

[GitHub](#)

PyTorch · MTCNN · FaceNet · SVM · Flask

- Engineered a real-time face recognition pipeline (MTCNN, FaceNet, SVM) deployed via a Flask web application for automated attendance logging.

• IMU-Coordinated Bot with Heading Stabilisation

[GitHub](#)

MPU6050 · Madgwick Filter · PID Control · Arduino

- Fused 6-DOF IMU data via Madgwick complementary filter; implemented closed-loop PID heading stabilisation with auto-calibration and deadband control.

• FemeVision — Vision-Based Safety System

[GitHub](#)

YOLOv8 · Lucas-Kanade Optical Flow · MediaPipe · OpenCV · Streamlit

- Multi-modal perception pipeline combining object detection, optical flow motion estimation, and pose estimation — Runner-up, Hack for Tomorrow Hackathon, VSSUT Burla 2025.

• Smart Environment Monitor (IoT)

[GitHub](#)

Raspberry Pi · Arduino Nano · ESP32 · ThingSpeak

- Multi-sensor RPi + Arduino hybrid system with real-time cloud telemetry via ThingSpeak; covers temperature, humidity, and ambient light acquisition.

Leadership & Achievements

- **Vice Chairperson — IEEE EDS Student Branch Chapter, PMEC** *2026 – 27*
 - Leading technical events, industry talks, and student outreach as elected Vice Chair of the IEEE Electron Devices Society Student Branch Chapter.
- **Coordinator R&I Wing — Robotics Club, PMEC** *2025 – 26*
 - Spearheading research and innovation initiatives; organised RoboExpo-2k25, Robomania 2k26, and the Innovation Challenge Hackathon.
- **Semifinalist — Robowar 8 kg, IIT KGP Kshitij 2026** *Jan 2026*
 - Designed and competed with an 8 kg combat robot nationally — hands-on embedded debugging and hardware-software integration under competition pressure.
- **Runner-up — Hack for Tomorrow Hackathon, VSSUT Burla** *Jan 2025*
 - Led 4-member team to build a real-time violence detection system using optical flow and Harris corner detection.
- **Student Volunteer — 1st IEEE Intl. Conference on AIMLCPS 2026, PMEC** *Feb 2026*
 - Managed hybrid research-paper sessions across 3 days as part of a 50-member volunteer team.

Certifications

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| • Autonomous Vehicles Bootcamp — Robolearn LLP | • Python for Data Science & ML — Udemy |
| • Generative AI: Learn by Doing — IIT Kharagpur | • Python Specialisation — University of Michigan |
| • Machine Learning A-Z — Udemy | • Embedded Systems for Beginners — NIELIT |

Research Interests

- **Autonomous Systems:** Robotics, Drone Simulations, Edge AI, Embedded Systems integration.
- **Artificial Intelligence:** Deep Learning, Computer Vision.

Extracurricular Activities

- **Technical Memberships:** Core Member at College Robotics Club and Active IEEE Student Member.
- **Interests & Hobbies:** Book reading, Traveling and Photography.